Acquisition of Phi-Features and Feature Inheritance in L1 English*

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Abstract

In the acquisition of English, children produce various types of errors that appear unrelated at first glance. Previous research has identified and discussed three such error types: optional infinitives (e.g., **Brian like wine*), erroneous case assignment (e.g., **Me like sushi*), and null subjects (e.g., *__ *like eggplant*). This study proposes that these errors arise during an intermediate stage in the acquisition of φ-features. Since the necessity of φ-features varies across languages, English-speaking children must acquire language-specific knowledge about their obligatory status in English. Under this view, learners are expected to pass through a stage in which their developing grammar lacks φ-features, resulting in non-target-like utterances. It is at this stage that the three error types are predicted to emerge.

1. Introduction

In the recent generative framework (e.g., Chomsky, 1995, 2013), Merge is regarded as the most fundamental operation, which combines two elements, α and β, into one constituent {α, β}. Merge is considered a basic property of language. Assuming that Merge is the only principle innate to humans, everything else, except for 3rd factors (Chomsky, 2005), must be acquired through linguistic experience. A crucial question arises here as to what children must acquire from linguistic input. One candidate for the answer is φ-features, as their necessity varies from language to language: some languages lack φ-features, while others require them. For instance, in English, C has φ-features [φ], whereas in Japanese, it does not (Saito, 2011). Thus, English-speaking children must come to know the existence and necessity of φ-features. Additionally, recent literature (e.g., Hayashi, 2020; Suenaga, 2024) suggests that feature inheritance is optional, raising a further question of whether feature inheritance is accurately applied, when

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necessary, in children's grammar. If English-acquiring children must learn the existence of φ-features and related operations, such as feature inheritance, through experience, then L1 English acquisition is expected to exhibit non-target-like production during the intermediate stages. Therefore, this study aims to explore production errors, particularly optional infinitives, erroneous Case assignment, and null subjects, committed by L1 English acquirers to investigate the acquisition of φ-features and feature inheritance.

2. Theoretical Background

2.1 Labeling Algorithm

Language minimally requires the operation Merge, which combines two elements to form a new syntactic object. Each syntactic object formed by Merge must have a label for interpretation at the sensorimotor and conceptual-intentional interfaces. According to Chomsky (2013, 2015), each label is determined by Labeling Algorithm (LA), which is an instance of Minimal Search (3rd factor), at the Transfer level.

When a set is formed by merging a head H with a phrase XP, H serves as the label, as illustrated in (1).

(1) $[_{\alpha}\ \mathrm{H}, \mathrm{XP}]$ $\alpha = \mathrm{H}$

When the set consists of two phrases, as in (2a), LA cannot identify the label in the same way as in case (1) because the heads of XP and YP are identified simultaneously. There are two possible ways to label a set like (2a). The first option is for one of the phrases to raise out of the set (i.e., Internal Merge), rendering the copy left behind invisible, leaving the remaining phrase as the only one visible to LA, as shown in (2b). The second solution involves the heads of the two phrases sharing agreement features, as in (2c), where a valued feature [F] on X and an unvalued feature [*u*F] on Y together provide the label as <F, F>.

(2) a. $[_{\alpha}\ \mathrm{XP}, \mathrm{YP}]$

b. $[\mathrm{XP}, \ldots [_{\alpha}\ \cancel{\mathrm{XP}}, \mathrm{YP}]]$ $\alpha = \mathrm{YP}$

c. $[_{\alpha}\ [\mathrm{X}_{[\mathrm{F}]}, \mathrm{WP}], [\mathrm{Y}_{[u\mathrm{F}]}, \mathrm{ZP}]]$ $\alpha = \langle \mathrm{F}, \mathrm{F} \rangle$

Let us now take a look at the derivation of a rudimentary sentence to illustrate how labeling operates.

(3) a. Mary likes John.

b. $[_{\delta}\ \text{Mary}_{[\varphi]}\ [_{\gamma}\ \text{v}_{[u\varphi]}\ [_{\beta}\ \text{John}_{[\varphi]}\ [_{\alpha}\ \text{R}, \cancel{\text{John}}]]]]$

c. $[_{\delta}\ \text{Mary}_{[\varphi]}\ [_{\gamma}\ \text{v}\ [_{\beta}\ \text{John}_{[\varphi]}\ [_{\alpha}\ \text{R}_{[v\varphi]}, \cancel{\text{John}}]]]]$ $\alpha = \text{R}, \beta = \langle\varphi, \varphi\rangle$

e. $[_{\delta}\ \text{Mary}_{[\varphi]}\ [_{\gamma}\ \text{R-v}\ [_{\beta}\ \text{John}_{[\varphi]}\ [_{\alpha}\ \cancel{\text{R}_{[v\varphi]}}, \cancel{\text{John}}]]]]$

f. $[_{\eta}\ \text{C}_{[u\varphi]}\ [_{\zeta}\ \text{Mary}_{[\varphi]}\ [_{\varepsilon}\ \text{T}\ [_{\delta}\ \cancel{\text{Mary}_{[\varphi]}}\ [_{\gamma}\ \text{R-v}\ [_{\beta}\ \text{John}_{[\varphi]}\ [_{\alpha}\ \cancel{\text{R}_{[v\varphi]}}, \cancel{\text{John}}]]]]$

g. $[_{\eta}\ \text{C}\ [_{\zeta}\ \text{Mary}_{[\varphi]}\ [_{\varepsilon}\ \text{T}_{[v\varphi]}\ [_{\delta}\ \cancel{\text{Mary}_{[\varphi]}}\ [_{\gamma}\ \text{R-v}\ [_{\beta}\ \text{John}_{[\varphi]}\ [_{\alpha}\ \cancel{\text{R}_{[v\varphi]}}, \cancel{\text{John}}]]]]$

$\gamma = \delta = \text{R-v}, \varepsilon = \text{T}, \zeta = \langle\varphi, \varphi\rangle$

h. $[_{\eta}\ \text{C}\ [_{\zeta}\ \text{Mary}_{[\varphi]}\ [_{\varepsilon}\ \text{T}_{[v\varphi]}\ [_{\delta}\ \cancel{\text{Mary}_{[\varphi]}}\ [_{\gamma}\ \text{R-v}\ [_{\beta}\ \text{John}_{[\varphi]}\ [_{\alpha}\ \cancel{\text{R}_{[v\varphi]}}, \cancel{\text{John}}]]]]$ $\eta = \text{C}$

The derivation of (3a) is outlined in steps (3b–h). First, R externally merge with *John*, which subsequently internally merges with α. Next, v and the external argument *Mary* are introduced by External Merge, reaching the phase level (3b). At the v phase, [*u*φ] on v is inherited by R via feature inheritance, and labels of the Transfer domain of v are determined (3c). Here, the unvalued φ-features on R agree with φ-features on DP (i.e., *John*). R then internally pair-merged with v (3e), thereby shifting phasehood from v to the lower copy of R, as v is invisible due to Pair-Merge (see Chomsky, 2004, for details on Pair-Merge). Since now phasehood is on the R, the complement of R, that is, the lower copy of *John*, is transferred. In (3f), T externally merges with δ, *Mary* internally merges with ε, and C externally merges with ζ, bringing the derivation to the C phase level. Like feature inheritance of v to R, [*u*φ] on C is inherited by T, after which LA determines the labels (3g). Here, I tentatively assume that the unvalued φ-feature on T is valued by feature agreement with the φ-feature on the DP in the Spec-TP (but see Section 2.4). Note that the TENSE-feature also originates in C (Chomsky, 2007, 2008) and is inherited with [*u*φ] by T, which is a crucial assumption in our subsequent discussion.

The sequence of phase-level operations can be summarized as follows.

(4) Feature inheritance → Labeling → Internal Pair-Merge (head movement) → Transfer

The order of phase-level operations described in (4), however, does not apply to the derivation of copula sentences where V-to-T movement is involved. Suppose that the derivation of copulas also follows the sequence (4).

(5) a. Mary is happy.

b. $[_{\delta}\ \text{C}_{[u\varphi]}\ [_{\gamma}\ \text{Mary}_{[\varphi]}\ [_{\beta}\ \text{T}\ [_{\alpha}\ \text{R-v}\ [\ldots]]]]]$

c. $[_{\delta}\ \text{C}\ [_{\gamma}\ \text{Mary}_{[\varphi]}\ [_{\beta}\ \text{T}_{[v\varphi]}\ [_{\alpha}\ \text{R-v}\ [\ldots]]]]]$ $\alpha = \text{R-v}, \beta = \text{T}, \gamma = \langle\varphi, \varphi\rangle$

d. $[_{\delta}\ \text{C}\ [_{\gamma}\ \text{Mary}_{[\varphi]}\ [_{\beta}\ \text{R-v-T}_{[v\varphi]}\ [_{\alpha}\ \cancel{\text{R-v}}\ [\ldots]]]]]$

e. $[_{\delta}\ \text{C}\ [_{\gamma}\ \text{Mary}_{[\varphi]}\ [_{\beta}\ \text{R-v-T}_{[v\varphi]}\ [_{\alpha}\ \cancel{\text{R-v}}\ [\ldots]]]]]$

The derivation of the sentence *Mary is happy* reaches the C phase (5b). [$u\varphi$] on C is transferred to T, and labels of the domain of C-complement (i.e., α, β, and γ) are determined in (5c). R-v is Pair-Merged with T, rendering the valued φ-feature on T invisible (5d). In (5e), the labeled complement of C is Transferred to the interface. What is problematic here is the invisible [$v\varphi$] on T. Since the feature is invisible in point where Transfer takes place, the sensorimotor interface has no choices to externalize the [$v\varphi$].[1]

To circumvent this issue, I posit that in the copula derivation, φ-features on C are inherited by R-v amalgam, instead of T, and the R-v is internally Pair-Merged before labeling. (6) represents the derivation of copula sentences.

(6) a. $[_{\delta}\ \mathrm{C}_{[u\varphi]}\ [_{\gamma}\ \mathrm{Mary}_{[\varphi]}\ [_{\beta}\ \mathrm{T}\ [_{\alpha}\ \text{R-v}\ [\ldots]]]]]$

b. $[_{\delta}\ \mathrm{C}\ [_{\gamma}\ \mathrm{Mary}_{[\varphi]}\ [_{\beta}\ \mathrm{T}\ [_{\alpha}\ \text{R-v}_{[u\varphi]}\ [\ldots]]]]]$

d. $[_{\delta}\ \mathrm{C}\ [_{\gamma}\ \mathrm{Mary}_{[\varphi]}\ [_{\beta}\ \text{R-v}_{[v\varphi]}\text{-T}\ [_{\alpha}\ \cancel{\text{R-v}_{[v\varphi]}}\ [\ldots]]]]]$ β = R-v-T, $\gamma = \langle\varphi, \varphi\rangle$[2]

e. $[_{\delta}\ \mathrm{C}\ [_{\gamma}\ \mathrm{Mary}_{[\varphi]}\ [_{\beta}\ \text{R-v}_{[v\varphi]}\text{-T}\ [_{\alpha}\ \cancel{\text{R-v}_{[v\varphi]}}\ [\ldots]]]]]$

The derivation is brought to the C phase level in (6a), and [$u\varphi$] is inherited by R-v in (6b) (cf. (5)). Before the labels of the domain of the C-complement are determined, R-v moves to form R-v-T. Crucially, the φ-features on R-v-T are visible because R-v, not T, bears the φ-features. To sum up, copula sentences involve long-distance feature inheritance of [$u\varphi$] from C to V, and the order of the phase-level operations varies depending on verb types (e.g., lexical vs. copula). When the verb is lexical, labeling precedes head-movement; when it is a copula, head-movement precedes labeling.

2.2 Labeling Weakness

In addition to LA, Chomsky (2015) posits that T in English and R in all languages are too weak to serve as the label. Although the concept of a weak head seems to elegantly reduce EPP and ECP effects (see Chomsky, 2015), several linguists have cast doubt on the notion (Goto, 2017; Hayashi, 2020; Mizuguchi, 2022, 2024; Messick, 2020; Shim, 2018, among others). Therefore, following Hayashi (2020) and Suenaga (2024), I assume that T and R are universally strong (see Hayashi, 2020, for a detailed argument for eliminating weak/strong distinctions).

An issue that needs to be addressed here is how ECM and EPP can be accounted for under the assumption of a strong T. In the remainder of this subsection, we will discuss

[1] If we assume that [$v\varphi$] is only invisible in syntax but visible at the interface (or externalization), the invisibility of [$v\varphi$], caused by Pair-Merge, might not be a problem. I leave this possibility open for future investigation.

[2] The label of α is open for question, but here I assume that α is determined by XP sister to R-v. In the case of (6), the AP headed by A (i.e., *happy*) serves as a label of α.

Hayashi's (2020) explanation of ECP effects, and in the subsequent section, we will highlight issues with this analysis. Section 2.4. provides a solution to this problem, from which the EPP effect is also deduced.

First, consider the following interrogative sentences where the *wh*-subject is extracted from the embedded clause.

(7) a. *Who$_i$ do you think that t_i loves Miki?
b. Who$_i$ do you think t_i loves Miki?

As seen in the contrast between (7a) and (7b), the movement of the *wh*-subject out of a *that*-clause is banned, while such movement is allowed without overt *that*. This well-known *that*-trace effect is deduced from the ECP in the Government and Binding framework (Chomsky, 1981). Within the current framework, Hayashi offers a more principled explanation in terms of labeling. The derivation of (8a) up to the embedded CP phase level follows either (8b) or (8c).

(8) a. *Who do you think that loves Miki?
b. $[_{\delta}\ \text{that}\ [_{\gamma}\ \text{who}_{[\varphi]}\ [_{\beta}\ \text{T}_{[v\varphi]}\ [_{\alpha}\ \cancel{\text{who}_{[\varphi]}}\ \text{loves Miki}]]]]$ $\alpha = \text{v}, \beta = \text{T}, \gamma = \langle\varphi, \varphi\rangle$
c. $[_{\varepsilon}\ \text{who}_{[\varphi]}\ [_{\delta}\ \text{that}\ [_{\gamma}\ \cancel{\text{who}_{[\varphi]}}\ [_{\beta}\ \text{T}_{[v\varphi]}\ [_{\alpha}\ \cancel{\text{who}_{[\varphi]}}\ \text{loves Miki}]]]]]$ $\alpha = \text{v}, \beta = \gamma = \text{T}$

In (8b), the subject *who* internally merges with β, and γ is labeled as <φ, φ>. Despite the unproblematic labeling, *who* in Spec-TP is unavailable in the next phase due to the Phase Impenetrability Condition (PIC). To escape the PIC, it must first move to the edge of the phase, that is, Spec-CP, as shown in (8c). In this case, Hayashi (2020) argues, γ cannot be labeled as <φ, φ>, which results in a violation of Full Interpretation since, he assumes, <F, F> functions as an assigner of a value to [*u*F] at the interfaces.

Next, let us look at the derivation of (9b).

(9) a. Who do you think loves Miki?
b. $[_{\delta}\ \text{C}\ [_{\gamma}\ \text{who}_{[\varphi]}\ [_{\beta}\ \text{T}_{[v\varphi]}\ [_{\alpha}\ \cancel{\text{who}_{[\varphi]}}\ \text{loves Miki}]]]]$ $\alpha = \text{v}, \beta = \text{T}, \gamma = \langle\varphi, \varphi\rangle$
c. $[_{\delta}\ \text{ø}\ [_{\gamma}\ \text{who}_{[\varphi]}\ [_{\beta}\ \text{T}_{[v\varphi]}\ [_{\alpha}\ \cancel{\text{who}_{[\varphi]}}\ \text{loves Miki}]]]]$

When the derivation reaches the CP phase, the labels of α, β, and γ are the same as in (8b). Since C is deleted, phasehood shifts to T, along with the shift of the Transfer domain to T's complement, thereby allowing *who* to participate in the next phase (see Chomsky, 2015).

2.3 Feature Inheritance

Chomsky (2007, 2008) proposes that tense and agreement features originate in C and must be inherited by T through feature inheritance, as seen in (3).

Feature inheritance in English provides a natural explanation for the differences in the interrogatives between English and Japanese within the labeling theory. In interrogative sentences, English involves T-to-C movement, whereas Japanese does not (at least overtly), as shown in (10) and (11). ('EA' stands for an external argument.)

(10) a. What does the boy read?

b. $[_{\mathrm{CP}}\ \textit{what}\ [_{\mathrm{CP}}\ \text{T-C}\ [_{\mathrm{TP}}\ \text{EA}\ [_{\mathrm{TP}}\ \cancel{\text{T}}\ [\ldots]]]]]$

(11) a. Otokonoko-wa nani-o yomi-masu-ka?

boy-TOP what-ACC read-POL-Q

'What does the boy read?'

b. $[_{\mathrm{CP}}\ [_{\mathrm{TP}}\ \text{EA}\ [_{\mathrm{TP}}\ [_{\mathrm{vP}}\ \ldots\textit{what}\ldots]\ \text{T}]]\ \text{C}]$

The distinct behavior of the head movement in interrogatives can be attributed to the presence or absence of φ-features. First, consider the detailed derivation of the English interrogative in (10a).

(12) a. $[_{\delta}\ \textit{wh}_{[u\text{Q}]}\ [_{\gamma}\ \text{C}_{[u\varphi][\text{Q}]}\ [_{\beta}\ \text{EA}_{[\varphi]}\ [_{\alpha}\ \text{T}\ [\ldots]]]]]$

b. $[_{\delta}\ \textit{wh}_{[u\text{Q}]}\ [_{\gamma}\ \text{C}\ [_{\beta}\ \text{EA}_{[\varphi]}\ [_{\alpha}\ \text{T}_{[u\varphi][\text{Q}]}\ [\ldots]]]]]$ $\alpha = \text{T}, \beta = \langle\varphi, \varphi\rangle$

c. $[_{\delta}\ \textit{wh}_{[u\text{Q}]}\ [_{\gamma}\ \text{T}_{[u\varphi][\text{Q}]}\text{-C}\ [_{\beta}\ \text{EA}_{[\varphi]}\ [_{\alpha}\ \cancel{\text{T}_{[u\varphi][\text{Q}]}}\ [\ldots]]]]]$ $\gamma = \text{T-C}, \delta = \langle\text{Q}, \text{Q}\rangle$

The operations in the derivation of the CP phase are as follows: (i) [*u*φ] is inherited by T and [Q] is pied-piped, (ii) α and β are labeled as T and <φ, φ>, respectively, (iii) T pair-merges with C, (iv) γ and *δ* are labeled as T-C and <Q, Q>, respectively. In (10), T-to-C movement is motivated by the labeling of *δ* as <Q, Q>. T with Q-feature must move to C; otherwise, the label of *δ* cannot be determined as <Q, Q>, resulting in a crash. To illustrate this point clearly, let us suppose that T remains in situ. In this scenario, since [Q] is not on C, but T, *δ* remains unlabeled, as a result of which, the sentence, such as (10a), cannot be interpreted as the interrogative. Simply put, T-to-C movement in English arises from the inheritance of φ-features from C to T and pied-piping of the Q-feature.[3] Japanese, on the other hand, lacks φ-features (e.g., Saito, 2011), which naturally accounts for the absence of T-to-C movement.

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[3] This analysis cannot capture the indirect questions such as (ia).

Hayashi (2020) proposes that feature inheritance is an optional operation, and only unproblematic derivations survive (see Hayashi, 2020, for a detailed discussion on the motivation for the optional feature inheritance). If feature inheritance is optional, a new problem arises: *wh*-subject in the embedded clause can escape the PIC domain, as seen in cases like (7a). Consider (13).

(13) a. *Who do you think that loves Miki?

b. $[_{\zeta}$ who$_{[\varphi]}$ $[_{\varepsilon}$ that$_{[u\varphi]}$ $[_{\delta}$ ~~who$_{[\varphi]}$~~ $[_{\gamma}$ T $[_{\beta}$ ~~who$_{[\varphi]}$~~ $[_{\alpha}$ R-v …]]]]]]

$\alpha = \beta =$ R-v, $\gamma = \delta =$ T, $\varepsilon =$ C, $\zeta = \langle\varphi, \varphi\rangle$

In (13), feature inheritance of [$u\varphi$] from C to T is not applied, and the [$u\varphi$] and *who* in Spec-CP agree, assigning the label of ε as <φ, φ>. Since *who* is at the edge of the phase, it is available to further derivation. In spite of the seemingly converging derivation, the derivation in (13b) faces a problem at the sensorimotor interface because, in English, a valued φ-feature on a given head must hop to the head immediately below through affix hopping. In (13b), the φ on C cannot hop to V due to the intervening T (see Hayashi, 2020; Suenaga, 2024). Because of the locality requirement of affix hopping, optionality in feature inheritance is not problematic in (13).

In an interrogative, inducing the ECM effect, with an embedded clause with a lexical verb such as *love* and *kick*, Hayashi's analysis encounters no issues, but when the sentence involves copulas or auxiliary verbs, inserted into or raised to T, his analysis faces a problem. Consider the following example of an interrogative.

(14) a. *Who do you think that was happy?

b. $[_{\zeta}$ who$_{[\varphi]}$ $[_{\varepsilon}$ that$_{[u\varphi]}$ $[_{\delta}$ ~~who$_{[\varphi]}$~~ $[_{\gamma}$ R-v-T $[_{\beta}$ ~~who$_{[\varphi]}$~~ $[_{\alpha}$ ~~R-v~~ …]]]]]]

$\alpha = \beta =$ R-v, $\gamma = \delta =$ T, $\varepsilon =$ C, $\zeta = \langle\varphi, \varphi\rangle$

Unlike (13b), (14b) involves V-to-T movement. This head movement is applied after labeling, resulting in the label of γ and δ as T, not R-v-T (see Chomsky, 2015). The φ on C can lower to T satisfying the locality requirement of affix hopping, and contrary to the fact, the derivation converges.

(i) a. I don't know what you ate.

b. *I don't know what did you eat.

If Q-feature always goes into V along with φ-feature inheritance, the (un)grammaticality of (i) is difficult to explain. I speculate that in (ia), R<KNOW> has Q-feature, which is inherited by the embedded C, agreeing with [uQ] on *what*. I leave this possibility and consequences for future investigation.

2.4 Valuation of [$u\varphi$] on C

In Hayashi (2020), an unvalued φ is assigned a value at the sensorimotor interface through the label $<\varphi, \varphi>$; that is, the label $<\varphi, \varphi>$ is required for [$u\varphi$] to receive a value. As a consequence, Internal Merge of *who* in Spec-CP is incorrectly licensed, as in (14b). However, if we assume that Minimal Search identifies [$u\varphi$] on C and [φ] on DP in Spec-TP, and that [$u\varphi$] is valued (e.g., Chomsky, 2019), then the relation between C and the DP must be local, which allows us to resolve the issue discussed in (14). This analysis reduces EPP effects to the locality of the probe-goal relation in terms of feature valuation. Section 4 provides further evidence for the analysis.

Let us look at the derivation of (14), repeated below.

(15) a. *Who do you think that was happy?

b. $[_{\zeta}\ \text{who}_{[\varphi]}\ [_{\varepsilon}\ \text{that}_{[\nu\varphi]}\ [_{\delta}\ \cancel{\text{who}_{[\varphi]}}\ [_{\gamma}\ \text{R-v-T}\ [_{\beta}\ \cancel{\text{who}_{[\varphi]}}\ [_{\alpha}\ \cancel{\text{R-v}}\ \ldots]]]]]]$

$\alpha = \beta = \text{R-v},\ \gamma = \delta = \text{T},\ \varepsilon = \text{C},\ \zeta = <\varphi, \varphi>$

Under our assumption, the overt argument must be in Spec-TP in order to value an unvalued φ on C when Minimal Search is applied; otherwise, the unvalued φ would remain unvalued, resulting in the ungrammatical sentence with an uninflected verb.[4] An important assumption is that copies are invisible in syntax; to put it in another way, a copy of *who* in (15b) is invisible to Minimal Search. Hence, the derivation (15b) is ruled out.

If this line of analysis is on the right track, we predict that languages lacking φ-features, such as Korean and Japanese[5] do not require an overt argument in Spec-TP. This prediction is borne out by the Japanese sentences below.

(16)	(Boku-wa)	banana-o	tabe-ta.
	I-TOP	banana-ACC	eat-PAST
	'I ate a banana.'		

As shown in (16), a subject does not need to be overtly present.[6]

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[4] If Minimal Search can be applied at the time that C is introduced (before *who* moves over C), this analysis does not work. Hence, we need to assume that Minimal Search is applied after a phase-domain completes.

[5] Some linguists assume that Japanese has unvalued φ-features on C despite the lack of externalization of them (e.g., Mizuguchi, 2024).

[6] Since the EPP effects are deduced from the locality requirement of probe-goal relation, Japanese without [$u\varphi$] is expected to have no EPP effects. See Fukui (1995) for the argument for the lack of the EPP in Japanese; see also Miyagawa (2001) and Koizumi and Tamaoka (2010) for the argument for the EPP feature in Japanese; see also Tazaki (submitted) for the argument for the EPP in unergative/unaccusative structures in Japanese.

One might wonder why languages like Italian, which has φ-features on C, allow null subjects. Under our assumption that overt Spec-TP is necessary for valuing an unvalued φ on C, null subjects should not be permitted. Nevertheless, sentences such as (17) are possible.

(17) Mangio una mela.
eat-PRES a-apple
'(I) eat an apple.'

The reason for the omission of the subject lies in the fact that, in Italian, v pair-merges with T. Goto (2017) suggests that Italian v inherently bears [φ] due to the rich agreement morphology of verbs. Building on Goto's argumentation, Hayashi (2022) assumes that [*u*φ] on T is appropriately interpreted at the sensorimotor interfaces because of Pair-Merge of v with [φ] and T, as illustrated in (18).

(18) a. R-v[φ] → R-v[φ]-T[*u*φ]

Since, different from Hayashi's (2020) assumption that feature-assignment is carried out at the interfaces, we assume that feature-valuation takes place in syntax, [*u*φ] on T is valued when R-v pair-merged with the T.[7] Hence, in Italian sentences without an overt subject, as in (18), [*u*φ] on C is inherited by T without valuation but is valued when R-v raises to T due to the inherent [φ] on v.

Taken together, although the analysis proposed here suggests that an unvalued φ is valued through the local relation between a head bearing [*u*φ] and an overt DP bearing [φ], feature inheritance is still assumed to be optional. I do not go into detail in this article, as this is irrelevant to our discussion (but see Hayashi, 2020, for the evidence for optional feature inheritance).

3. L1 English Acquisition

This section discusses errors observed in the early L1 acquisition of English and suggests that these errors are expected under the current framework outlined in the preceding section. We focus particularly on well-documented errors concerning verb inflections, Case, and null

[7] Another possibility is that Pair-Merge of R-v with T renders T invisible along with [*u*φ] so that [*u*φ] is not problematic, regardless of whether it is valued or not. I leave this question open for further studies.

subjects and demonstrate that these errors can be uniformly accounted for in terms of φ-features and labeling failure.

3.2. Optional Infinitive

In the production of English-speaking children aged roughly from the end of the first year to the beginning of the third year, verbs are often produced uninflected, as exemplified in (19). In sentences where a verb must be inflected, the verb is frequently produced uninflected in child English grammar. (19a–b) are taken from Guasti (2016, taking examples from CHIDLES (MacWhinney & Snow, 1989; Brown, 1973)), and (19c–e) from Radford (1990).

(19) a. Mumma ride horsie. (Sarah, 2;6)
b. Marie go. (Sarah, 2;3)
c. Hayley draw it. (Hauley 20)
d. Man go up there. (Claire 23)
e. Mummy cry. (Jem 23)

In (19), bare infinitive forms are used in a finite context. Children at the stage of Optional Infinitive (OI) also produce verbs correctly inflected for tense and agreement. Furthermore, copulas are almost always inflected correctly.

Galasso (1999) examined natural production of an English-speaking child (Nicolas) and found a stage prior to the OI stage. He observed that Nicolas did not produce 3rd person singular -*s* until the beginning of the third year. As shown in Table 1, at the age of between 2;3 and 3;1, no instances of the -*s* form were found. Simply put, Nicolas at the age of from 3;2 to 3;6 is in the OI stage.

Table 1: Development of inflection: occurrence in obligatory contexts (based on Table 5.6 in Galasso, 1999, p. 106)

Age	3sg-pres + *s*
2;3–3;1	0/69
3;2–3;6	72/168 (43%)

OIs are also found in negation sentences, as illustrated in (20) (adopted from Guasti & Rizzi, 2002).

(20) a. Robin don't play with pens. (Adam 28, 3;4)
b. So Paul doesn't wake up. (Adam 28, 3;4)

As (20) shows, children optionally use infinitive form even with negation with *do*-support. Guasti and Rizzi (2002) and Sugisaki (2016) observe that *do* is almost perfectly inflected when it occurs in the interrogative, as in (21).

(21) Does dis write? (Adam 28, 3;4)

Sugisaki (2016) analyzed the corpora of 12 English-speaking children from CHILDES and indicated that OIs do not afflict *do* in the interrogative. The results of his analysis are presented below in Table 2. Table 2 includes data from eight children out of 12, four of whom were excluded because they produced few interrogatives with *do* or few errors in negation sentences. The total presented in the table was calculated by Dansako (2018, p. 345).

Table 2: The proportion of (un)inflected *do* (Dansako, 2018, p. 345)

	Negation		Interrogative	
	doesn't	**don't*	*does*	**do*
Total	150 (48.5%)	159 (51.4%)	228 (98.2%)	4 (1.7%)

3.2. Erroneous Case Marking

During the OI stage, children also make errors regarding Case assignment, as exemplified in (22). Children at this stage produce non-NOM subjects. (22a–b) are adopted from Galasso (1999).

(22) a. Me get it (Nicolas 21, 3;0)
b. Him eat (Nicolas 25, 3;6)

What is noticeable is that when non-NOM subject errors emerge, verbs are, more often than not, infinitive forms (e.g., Schütze & Wexler, 1996). Schütze & Wexler (1996) counted subject forms using data from CHILDES and quantitively show that non-NOM subjects are used more often with nonfinite verbs, as indicated in Table 3.

Table 3: Finiteness versus case (Nina, 3sg) (Schütze & Wexler, 1996, p. 674)

	Verb form	
Subject	+Finite	–Finite
NOM	225	139

Non-NOM	14	120
% of non-NOM	5%	46%

$X^2 = 115.7$, $p < .000001$

Intriguingly, Schütze & Wexler (1996) also suggest that in the case where verbs are inflected for past, but not for agreement with subjects, children produce non-NOM subjects, as given in Table 4. Although the number of instances is relatively small, the propensity requires explanation.

Table 4: Distribution by verb type (Nina & Sarah, 3sg) (based on Tables 6 and 11in Schütze & Wexler, 1996, pp. 674–676)

Child	Age	Verb form	NOM	Non-NOM
Nina 01–31	1;11.16–2;5.28	Main V with -*s*	10	0
		Past Tense	16	6
Sarah 26–46	2;8.25–3;1.24	Main V with -*s*	3	0
		Past Tense	7	3

3.3. Null Subject

Along with the OI phenomenon, another child production error has caught researchers' attention, which is null subject, as in (23) (Hyams, 1986, 2011; Hyams & Wexler, 1993; Orfitelli & Hyams, 2012; Radford, 1990; Rizzi, 1994; among others). Examples (23a–e) are taken from the speech of Eric at 20–22 months provided in Bloom (1970).

(23) a. Play it
b. Eating cereal
c. Shake hands
d. See window
e. No go in

Interestingly, children in the stage of null subjects (NS) sometimes restate the sentence with subjects after they produced a subjectless sentence, as shown in (24) (Braine, 1973; Hyams, 1986). (24a–c) are adopted from Braine (1973, as cited in Hyams, 1986) and (24d–e) are from Hyams (1986).

(24) a. close radio…Mommy close radio

b. take off…Daddy take off…Mommy take of

c. stand up…cat stand up (a–c from Johnathan, 26–27 mos.)

d. fall…stick fall

e. go nursery…Lucy go nursery

f. push Stevie…Betty push Stevie (d–f from Stevie, 25–26 mos.)

Such instances as (24) indicate that children who drop subjects know that subjects are necessary in English.

As one might noticed in the examples in (23) and (24), when NSs are produced, their verbs are highly likely to be nonfinite (Bromberg & Wexler 1995; Guasti, 2016; Schütze & Wexler, 2000; Wexler, 1998). For example, Schütze and Wexler (2000) recruited English-speaking children divided into three groups based on their ages (Young: 2;2–2;10; Mid: 3;0–3;5; Old: 3;6–3;11) and conducted an elicited production task to examine Optional Infinitives and NSs. Results of their study are summarized in Table 1.

Table 5: Rates of NSs with (un)inflected verbs (Schütze and Wexler, 2000, p. 675)

	Subject					
	Yong (N = 9)		Mid (N = 9)		Old (N = 7)	
Verb form	Overt	Null	Overt	Null	Overt	Null
Inflected	39	8 (17%)	29	17 (37%)	67	6 (8%)
Uninflected	39	40 (51%)	19	43 (69%)	14	47 (77%)

As Table 5 suggests, NSs are observed much more when the verbs are uninflected. Despite the fact that majority of NSs appear with uninflected lexical verbs, NSs even with inflected verbs have also been reported in the literature, as will be seen (see also Guasti, 2016 and Phillips, 2010).

Moreover, children seldom omit subjects when verbs are copula (Sano & Hyams, 1994). Sano and Hyams (1994) counted the number of NSs with the uncontracted *am*, *are*, *is* in the corpora of Eve, Adam, and Nina (CHIDLES; MacWhinney & Snow, 1989; Brown, 1973; Suppes, 1974).

Table 6: The proportion of NSs with *am, are, is* (adopted from Table 1 in Sano & Hyams (1994, p. 548))

File	Age	*am*	*are*	*is*
Eve 01–20	1;6–2;3	0/4	0/36	0/109

Adam 01–20	2;3.4–3;0.11	0/1	0/81	13/114 (11.4%)
Nina 01–21	1;11.16–2;4.12	0/0	0/19	2/50 (4%)

Note. NINA 08 is not available, hence NINA 01–21 consists of 20 files.

Table 6 shows that NSs are very infrequent with copulas. To compare the number of NSs with copulas to the number of NSs with non-copulas and non-auxiliaries, Sano and Hyams summarize subjectless sentences with lexical verbs reported in Hyams and Wexler (1993) and Pierce (1992). The proportion of NSs with lexical verbs is displayed in Table 7 (adopted from Table 2 in Sano & Hyams, 1994, p. 549).

Table 7: The proportion of NSs with lexical verbs (i.e., non-copulas, non-auxiliaries)

Child	Age	Proportion
Eve	1;6–2;1	26%
Adam	2;5–3.0	41%
Nina	1;11.16	44%
	2;26	11%

As seen in Table 7, children are more likely to use NSs when the verb is lexical than when it is a copula.

Sano and Hyams also investigated NSs with verbs inflected with past -*ed* and 3rd person singular -*s*. Tables 8 and 9 exhibit the proportion of NSs with verbs with -*ed* and -*s*, respectively.

Table 8: NSs with verbs inflected with -*ed* (adopted from Table 4 in Sano & Hyams (1994, p. 550))

File	Age	Proportion
Eve 01–20	1;6–2;3	22.5% (9/40)
Adam 01–20	2;3–3;0	56.4% (13/23)
Nina 13–21	2;2–2;4	18.8% (3/16)

Table 9: NSs with verbs inflected with -*s* (adopted from Table 6 in Sano & Hyams (1994, p. 552))

File	Age	Proportion
Eve 01–20	1;6–2;3	10% (5/50)
Adam 01–20	2;3–3;0	25.8% (16/62)

Tables 6 and 7 compared, the differences in the proportion of NSs between sentences with copulas and lexical verbs inflected for past can be seen: NSs are more frequent with the morpheme *-ed* than with copulas. Noticeably, Tables 8 and 9 indicates that children production of NSs with verbs inflected with *-s* occur less often than those with *-ed* but more often than those with copulas. The frequency of NSs can be summarized as below ('A > B' means 'A occur more frequently than B').

(25) Lexical without inflection > Lexical with *-ed* > Lexical with *-s* > Copula

There is another tendency with regard to NSs, which is that pronouns seem to be dropped more frequently than lexical verbs (e.g., Gerken, 1991; Hyams & Wexler, 1993). Hyams and Wexler's (1993) analysis of Adam and Eve's production in CHILDES shows that as the children grow older, the proportion of pronominal subjects increases, while the number of lexical subjects, more or less, remains the same. Take Adam's files. Adam 06 has 11% of pronominal subjects and 33% of lexical subjects, whereas Adom 30 has 67% of pronominal subjects and 30% of lexical subjects. On the basis of these findings, Hyams and Wexler suggest that at the early stage of acquisition (around the OI stage), NSs are prone to be produced in a sentence where non-null subject languages use pronouns, and that as the proportion of NSs decreases, the proportion of pronominal subjects increases.

3.4. Summary

In this section, we reviewed three error types, namely, optional infinitive, non-NOM subject, and null subject. The error patterns are summarized below in (24). The following section discusses how these errors are explained within the current theory.

(26)
a. Infinitive forms are optionally used in the context where the 3rd person singular *-s* is obligatory.
b. Infinitive forms are optionally used in negative sentences with *do*, but almost never occur in interrogative sentences with *do*
c. Non-NOM subjects are used more frequently with uninflected verbs than with inflected verbs.
d. Non-NOM subjects are used more frequently with verbs inflected with *-ed* than with those inflected with *-s*.
e. NSs occur more with nonfinite verbs than with finite verbs.
f. NSs occur in the following proportion:
Lexical without inflection > Lexical with *-ed* > Lexical with *-s* > Copula

4. Early L1 English Errors as a Consequence of Unacquired Phi-Features

In Section 3, we discussed three often-found errors, namely, NSs, OIs, and erroneous Case marker. The present section explores the acquisition of the target-like φ-features and the feature-related operation. It also demonstrates that incomplete acquisition of φ-features accounts for all the errors summarized in (26).

The start of our discussion about the acquisition of φ-features is cross-linguistic differences in terms of the presence or absence of φ-features. Languages like English clearly have φ-features, as they show subject-object agreements, while languages like Japanese lack φ-features (see Section 2.4). Children thus need to figure out the presence or absence of φ-features from experience (see also Murasugi, 2020). In this case, there is a huge possibility that child grammar undergoes the stage where the necessity of φ-features in English is not yet acquired. We can also predict that child English grammar at this stage do not show the subject-object agreement. The prediction is borne out by Galasso's (1999) analysis of Nicolas' natural production that shows that English-speaking children go through a stage where all verbs are uninflected, as seen in Section 3.1.[8]

In the OI stage, children begin to acquire the existence of φ-features in English. In other words, during this stage, the child grammar is in the middle of the full acquisition of φ-features and thus shows uninflected verbs. I arbitrarily assume here that the TENSE-feature is always present in child grammar during the OI phase, but we may reveal that it also exhibits optionality in the future. We leave this question open for future scrutiny. Since 3rd person singular *-s* is the realization of [φ], without φ-features, uninflected verbs are naturally expected. In addition, even if [φ] is on C, a verb does not inflect unless the [φ] is inherited by T. If feature inheritance does not take place, the verb will not be inflected due to the intervening T, which prevent affix to hop into V (R-v), as illustrated in (27).

(27) $[_{\gamma}\ \mathrm{C}_{[v\varphi]}\ [_{\beta}\ \mathrm{EA}_{[\varphi]}\ [_{\alpha}\ \mathrm{T} \ldots]]]$[9]

(26a) is now explained. The OI phenomena happen because the derivation lacks either φ-features on C or feature inheritance.

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[8] Some researchers (e.g., Galasso, 1999; Murasugi, 2020; Radford, 2000) assume that children at this stage maximally have vP without functional categories such as T and C. I do not discuss this stage and focus only on the OI stage in this paper (see Galasso, 1999, and Murasugi, 2020, for a detailed discussion).

[9] In the examples in this section, I indicate the existence of [TENSE] only when necessary.

This line of analysis also captures the optionality observed in negative sentences such as (20), repeated below in (28).

(28) a. Robin don't play with pens. (Adam 28, 3;4)
b. So Paul doesn't wake up. (Adam 28, 3;4

Target-like negative sentence (28b) is produced in a target-like way, as shown in (29). (30a) are associated with (28a). (30b, c) are other possibilities. (For the sake of simplicity, I do not indicate EA in Spec-v)

(29) $[_{\varepsilon}$ C $[_{\delta}$ $EA_{[\varphi]}$ $[_{\gamma}$ $T_{[v\varphi][TENSE]}$ $[_{\beta}$ *not* $[_{\alpha}$ R-v …]]]]

(30) a. $[_{\varepsilon}$ C $[_{\delta}$ $EA_{[\varphi]}$ $[_{\gamma}$ $T_{[TENSE]}$ $[_{\beta}$ *not* $[_{\alpha}$ R-v …]]]] no [φ], inheritance of [TENSE]
b. $[_{\varepsilon}$ $C_{[v\varphi][TENSE]}$ $[_{\delta}$ $EA_{[\varphi]}$ $[_{\gamma}$ T $[_{\beta}$ *not* $[_{\alpha}$ R-v …]]]] [φ], no inheritance
c. $[_{\varepsilon}$ $C_{[TENSE]}$ $[_{\delta}$ $EA_{[\varphi]}$ $[_{\gamma}$ T $[_{\beta}$ *not* $[_{\alpha}$ R-v …]]]] no [φ], no inheritance

In (29), features are inherited by T. Since the features cannot hop to R-v due to the intervening head (*not*), *do* is inserted in T. *Do* is inflected as with adult grammar. (30a) does not incorrectly have φ-features on C, but the TENSE-feature is inherited by T. For the same reason for *do*-support in (29), [TENSE] is externalized as *do*. *Do* in (30a) cannot be inflected for agreement due to the absence of φ-features. (30b) and (30c) show the identical externalization despite the differences: the former has φ-features on C, while the latter does not. Neither of the derivations involves feature inheritance. The lack of feature inheritance (30b, c) disallows *do*-support because of the auxiliary nature of *do* (Hayashi, 2020). These structures have been occasionally documented, as in (31).

(31) a. He no bite you
b. I no want envelop
c. I no taste them (a–c from Klima & Bellugi (1966), as cited in Wexler (1994: 330)
d. Daddy not mend his car. (Neville, 30 months)
e. That not go on there. (Robert, 33 months)
f. Mummy not have tea. (Sheila, 42) (d–e from Radford 1994)

Examples in (31) can be taken as evidence that (30b) and (30c) are what happens in child grammar.

To fully explain (26b), we are also required to give an explanation of non-occurrence of the uninflected *do* in the interrogative. This phenomenon can be captured by pied piping of the Q-feature. Recall that the Q-feature may be inherited by T along with φ-features from C and that if φ-features do not exist on T, T-to-C movement does not occur (see Section 2.3). Given that argument, *do* that appears in the interrogative in child production indicates that φ-features are present on C and inherited by T. If this is on track, when *do* appears in the interrogative, it must be inflected. One might wonder whether feature inheritance is applied to Q-feature when it does to TENSE-feature, even if φ-features are absent and consequently, uninflected *do* is inserted in T-C after the head movement. This is not true, as Q-feature is not pied piped with TENSE-feature but only with φ-features, as we saw in Section 2.3. Taken together, the occurrence of *don't* and *doesn't* in child grammar during the same period and the non-occurrence of uninflected *do* in the interrogative are accounted for by the presence or absence of φ-features.

We next turn to erroneous use of non-NOM subjects. In Section 3.2, we saw that children in the OI stage produce more non-NOM subjects with uninflected verbs, in comparison with those with inflected verbs (i.e., (26c)). Use of nonfinite verb forms in child grammar implies the absence of φ-features. Important premises in the discussion here are that an unvalued φ on C must be valued by φ on DP in the local domain, that is, in Spec-TP (see Section 2.4), and that DP receives Nominative Case in a position c-commanding T, that is, Spec-TP. With these assumptions in mind, let us take a look at the derivation without φ-features.

(32) a. $[_{\delta}\ \mathrm{C}\ [_{\gamma}\ \mathrm{T}\ [_{\beta}\ \mathrm{EA}_{[\varphi]}\ [_{\alpha}\ \text{R-v} \ldots]]]]$
b. $[_{\varepsilon}\ \mathrm{C}\ [_{\delta}\ \mathrm{EA}_{[\varphi]}\ [_{\gamma}\ \mathrm{T}\ [_{\beta}\ \cancel{\mathrm{EA}_{[\varphi]}}\ [_{\alpha}\ \text{R-v} \ldots]]]]]$

Both (32a) and (32b) lack φ-features on C, which means that EA does not have to raise to Spec-TP. When (32b) is derived, Nominative Case is assigned to EA, but when (32a) is derived, a non-Nominative Case is assigned. In the case where C has φ-features, EA must move to Spec-TP to value an unvalued φ on C, showing no non-NOM subjects. In the same vein, the occurrence of non-NOM subjects with verbs inflected for past (i.e., (26d)) is explained. Considering TENSE-feature does not need to be valued by DP, unlike an unvalued φ, DP is not required to occupy Spec-TP. If this is the case, whether verbs are inflected for past tense has nothing to do with non-NOM subjects.

The remaining question is how NSs (e.g., (26e, f)) are captured. Thus far, I have touched upon the labeling of φ-less derivation. φ-features play an essential role in the labeling of [XP, YP] structures. Then, the derivation without φ-features on C should crash due to the unlabeled object, which needs to be resolved. I posit that NSs are the solution child grammar adapt. Look at the derivation without φ-features again.

(33) $[_\gamma\ \mathrm{C}\ [_\beta\ \mathrm{EA}\ [_\alpha\ \mathrm{T} \ldots]]]$ $\qquad$ $\alpha = \mathrm{T},\ \beta = ??,\ \gamma = \mathrm{C}$

Sets α and γ are in the [H, XP] configurations and labeled without trouble. The label of β, however, cannot be determined due to the [XP, YP] configuration. Without φ-features on C, there is no ways to label β. If we assume that children know that unvalued features must be valued to avoid a violation of Full Interpretation (Wexler, 1998), children circumvent the crashing derivation by 'deleting' the EA, so that β is labeled as T, as schematized in (34).[10,11]

(34) $[_\gamma\ \mathrm{C}\ [_\beta\ \mathbb{EA}\ [_\alpha\ \mathrm{T} \ldots]]]$ $\qquad$ $\alpha = \beta = \mathrm{T},\ \gamma = \mathrm{C}$

This is the reason for NSs. The deleted argument is interpreted from discourse contexts, which leads to the observation that many NSs are pronouns (see Section 3.1). If the deleted argument cannot be interpreted, the derivation crashes due to the violation of recoverability condition. Therefore, when the subject is new information that cannot be recovered from discourse contexts, the derivation (34) cannot converge: it must violate either Full Interpretation or the recoverability condition. This is probably the reason why overt subjects are still used even when φ-features are absent. It is important to note here that the use of NSs is the last resort in child English grammar; that is, when the derivation converges with overt subjects, the subjects will not be deleted.

Finally, we tackle the tendency of NSs in terms of verb types, as summarized below.

(35) Lexical without inflection > Lexical with -*ed* > Lexical with -*s* > Copula

Let us start with NSs with nonfinite verbs. This type is found in the following situations:

(36) a. $[_\varepsilon\ \mathrm{C}_{[\text{TENSE}]}\ [_\delta\ \mathbb{EA}_{[\boldsymbol{\varphi}]}\ [_\gamma\ \mathrm{T}\ [_\beta\ \cancel{\mathrm{EA}_{[\varphi]}}\ [_\alpha\ \text{R-v} \ldots]]]]$ $\qquad$ no [φ], no inheritance

b. $[_\varepsilon\ \mathrm{C}\ [_\delta\ \mathbb{EA}_{[\boldsymbol{\varphi}]}\ [_\gamma\ \mathrm{T}_{[\text{TENSE}(-\text{past})]}\ [_\beta\ \cancel{\mathrm{EA}_{[\varphi]}}\ [_\alpha\ \text{R-v} \ldots]]]]$ $\qquad$ no [φ], inheritance

c. $[_\varepsilon\ \mathrm{C}\ [_\delta\ \mathbb{EA}_{[\boldsymbol{\varphi}]}\ [_\gamma\ \mathrm{T}\ [_\beta\ \cancel{\mathrm{EA}_{[\varphi]}}\ [_\alpha\ \text{R-v}_{[\text{TENSE}(-\text{past})]} \ldots]]]]$ $\qquad$ no [φ], LDFI

d. $[_\varepsilon\ \mathrm{C}_{[u\varphi][\text{TENSE}]}\ [_\delta\ \mathbb{EA}_{[\boldsymbol{\varphi}]}\ [_\gamma\ \mathrm{T}\ [_\beta\ \cancel{\mathrm{EA}_{[\varphi]}}\ [_\alpha\ \text{R-v} \ldots]]]]$ $\qquad$ no inheritance

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[10] Since the derivation lacks unvalued φ-features, EA does not have to raise to Spec-TP. Even if this is the case, EA would be deleted due to the same labeling failure as (34).

[11] I do not go into the detail of the status of deleted arguments. See Rizzi (1994) and Radford (1990, 1994) for a relevant discussion.

This type of NSs is produced in the three structures above: (36a) without φ-features + feature inheritance, (36b) without φ-features + inheritance of TENSE-feature (–past), without φ-features + long-distance inheritance (LDFI) of TENSE-feature (–past), and (36d) with φ-features + without feature inheritance. In all the cases, the labels of δ cannot be determined with overt subjects, and the subjects are deleted.

Secondly, the possible derivation of NSs with verbs inflected with *-ed* is given in (37).

(37) a. $[_{\varepsilon}\ \mathrm{C}\ [_{\delta}\ \mathbb{EA}_{\boldsymbol{[\varphi]}}\ [_{\gamma}\ \mathrm{T}_{[\mathrm{TENSE}]}\ [_{\beta}\ \cancel{\mathrm{EA}_{[\varphi]}}\ [_{\alpha}\ \text{R-v} \ldots]]]]]$ no [φ], inheritance

b. $[_{\varepsilon}\ \mathrm{C}\ [_{\delta}\ \mathbb{EA}_{\boldsymbol{[\varphi]}}\ [_{\gamma}\ \mathrm{T}\ [_{\beta}\ \cancel{\mathrm{EA}_{[\varphi]}}\ [_{\alpha}\ \text{R-v}_{[\mathrm{TENSE}]} \ldots]]]]]$ no [φ], LDFI

c. $[_{\varepsilon}\ \mathrm{C}\ [_{\delta}\ \mathbb{EA}_{\boldsymbol{[\varphi]}}\ [_{\gamma}\ \mathrm{T}\ [_{\beta}\ \cancel{\mathrm{EA}_{[\varphi]}}\ [_{\alpha}\ \text{R-v}_{[v\varphi][\mathrm{TENSE}]} \ldots]]]]]$ [φ], LDFI

TENSE-feature is on T or R-v and thus is externalized in R-v. However, because the label of γ cannot be determined without φ on T, the subject EA is deleted.

Thirdly, NSs with *-s* only have the following structure.

(38) $[_{\varepsilon}\ \mathrm{C}\ [_{\delta}\ \mathbb{EA}_{\boldsymbol{[\varphi]}}\ [_{\gamma}\ \mathrm{T}\ [_{\beta}\ \cancel{\mathrm{EA}_{[\varphi]}}\ [_{\alpha}\ \text{R-v}_{[v\varphi][\mathrm{TENSE}]} \ldots]]]]]$ $\alpha = \beta = \text{R-v},\ \gamma = \delta = \mathrm{T},\ \varepsilon = \mathrm{C}$

In (38), a valued φ is inherited by R-v, instead of T, and δ cannot be labeled, resorting to the null subject. I assume that this long-distance feature inheritance possibly happens in NS sentences with verbs inflected with *-ed* as well.

I speculate that the long-distance feature inheritance is applied due to the input in which [*v*φ] is typically externalized in R-v except for sentences with copulas, auxiliary, and *do*-support (another reason for long-distance feature inheritance is also discussed). Simply put, there are two options to attach [*v*φ] to R-v in the case of lexical verbs.

(39) a. Affix hopping at the sensorimotor interface

b. Long-distance feature inheritance from C to V in syntax

Finally, we attempt to address the scarcity of NSs with copulas. As seen in Section 3.3, when the verb is a copula, its subject is rarely dropped. The resistance to NSs appears to suggest the successful acquisition of φ-features in English. Recall that the derivation, including copulas, involves a long-distance feature inheritance. Given that children fail to produce sentences with inflected lexical verbs, they are likely to be in the middle of the acquisition of the obligatoriness of [*v*φ] on T in the sentence including lexical verbs, but in the stage of the complete acquisition of the obligatoriness of [*v*φ] on V in the sentence including copulas. The successful acquisition of the obligatoriness of [*v*φ] in the copula sentences can be achieved due to the salience of the inflection of *be* in the input. Compared with lexical verbs, *be*-verbs have rich inflection, as illustrated below.

Table 10: Subject-Verb Agreement Inflection

Subject	Lexical	Copula
I	love	am
You	love	are
We	love	are
They	love	are
She/he	love**s**	is
Miki	love**s**	is

What is more, copulas are externalized as *be* when they appear after an auxiliary verb or an infinitive *to*. The distinct appearance of *be,* depending on the surrounding contexts, functions as evidence that English copulas must have [*v*φ]. With the help of the evidence observed from the input, children easily acquire the derivation of the copula sentences.

Children's success in producing target-like copula sentences indicates that in their grammar, [*v*φ] on C is correctly inherited by V (i.e., long-distance feature inheritance). This target-like long-distance feature inheritance is probably the cause of the non-target-like long-distance feature inheritance in sentences including lexical verbs (see (36c), (37b–c), and (38)).

The propensity of NSs (35) can be explained by the number of possible derivations for each type of verb and inflection. When a verb is uninflected, there are four possible derivations of an NS. When that is inflected with -*ed*, there are three possible derivations, and when that is inflected with -*s*, only one possible derivation. In the case of the verb being a copula, NSs rarely occur.

5. Conclusion

Through this article, we have reviewed the labeling algorithm and feature inheritance and examined whether these theories accommodate varying errors made by L1 English children. Some of the errors are elegantly explained, I think, but there are a number of things to account for and to provide empirical evidence for. Also, what is more important in my future development of this study is to bring more attention to the predictions derived from the analyses I proposed in this paper. I hope my study will give a new insight into the field of L1 acquisition and of linguistic theory.

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